

2. (a) Stop heating and stir the water well.

1A
1A 2

(b) (i) $\frac{L - 64}{80 - 64} = \frac{65 - 20}{92 - 20}$
 $L - 64 = 10$
 $L = 74 \text{ mm}$

1M

1A 2

- (ii) Let x °C be the absolute zero.

$$\frac{20 - x}{92 - 20} = \frac{64 - 0}{80 - 64}$$

$$20 - x = 288$$

$$x = -268 \text{ °C}$$

1M

1A 2

1. (a) Fair test (otherwise the initial temperature would affect the result).

1A

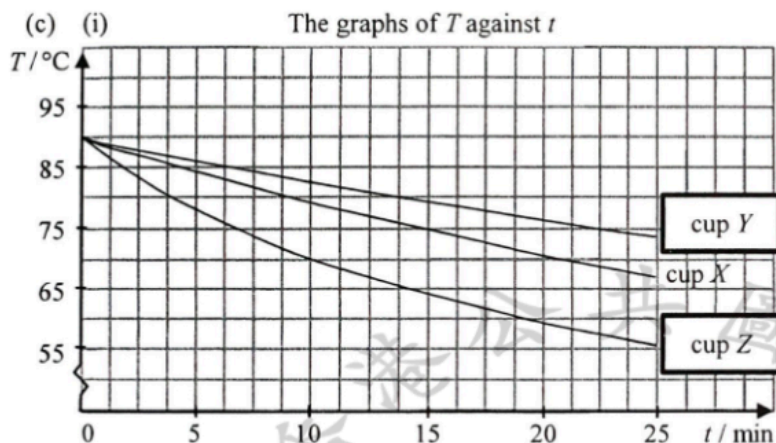
1

- (b) Temperature difference between hot water and the surroundings (room temperature) decreases as time elapses.
Rate of energy loss / heat loss becomes smaller,
hence rate of temperature drop is smaller and the curve becomes less steep.

1A

1A

2



1A

1

- (ii) Comparing to cup X as a control,
Cup Y :
The shiny surface of aluminium foil is a poor emitter /
good reflector of radiation (thus reduces heat lost to
surroundings).
Cup Z :
No lid allows convection through air / surroundings.

1A

1A

1A

Or
No lid enhances heat lost by evaporation / convection /
contacting with air / surroundings.

1A

3

- (d) Polystyrene / foam plastic or foam / wood

1A

1